

FROZEN SITE-PROBE ACCURACY IS NOT A STANDALONE HARMONIZATION SCORE: AN AUDIT OF EVALUATION PRACTICE FOR MULTI-SITE MRI

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ABSTRACT

Image-level harmonization of multi-site MRI is often evaluated with a site probe, a classifier that predicts the acquisition site: if its accuracy drops on harmonized images, site effects are considered removed. The probe is often trained once on original images and applied unchanged (a frozen probe). We test how reliable this verdict is on ABIDE I (1,112 T1-weighted scans, 17 sites), using ten outputs of seven harmonization methods. First, the verdict depended on how the probe was trained: retraining the frozen probe’s recipe with new random seeds moved its balanced accuracy (chance: 0.06) on one CycleGAN output between 0.13 and 0.59, and better-trained probes found more site information than the frozen probe in eight of the ten outputs. Second, a Gaussian blur, which moves images away from the target site, lowered the frozen probe’s accuracy to chance level, more than any harmonizer. Third, probes trained on the harmonized images themselves recovered the site far better than probes trained on original images, even from intensity histograms alone. The HACA3 harmonizer, for example, kept probes trained on original images near 0.28, yet its intensity histograms still revealed the site at 0.71. Low frozen-probe accuracy therefore does not show that site information was removed. We recommend reporting several well-trained probes, probes trained on the harmonized images, and alignment with the target site.

Index Terms— MRI harmonization, evaluation, site classification, domain shift, reproducibility

1. INTRODUCTION

Differences between scanners and protocols confound pooled analyses of multi-site MRI and degrade models on unseen sites [1, 2]. Image-level harmonization addresses this by translating scans toward a common appearance. Whether it succeeds is hard to verify: direct checks need travelling subjects [3] or paired acquisitions [4], which exist for only

a few scanners, so large retrospective cohorts are evaluated indirectly [5].

The most common indirect check is a *site probe*, a classifier trained to predict the acquisition site, a task that is easy on raw MRI [2]. If its accuracy drops on harmonized images, site effects are considered removed [6, 7]. Some studies retrain the probe on harmonized images [6] or image features [7, 8]; others train it once on original images and apply it unchanged, which we call a *frozen probe*. PRISM [9], for example, reports such a drop, with target-site predictions and segmentation checks, as evidence of harmonization. Probe accuracy is usually reported next to similarity to the input, such as PSNR.

These checks can mislead. A frozen probe may fail simply because harmonized images look unfamiliar, not because site information is gone. Removing an attribute until a fixed classifier fails does not prevent a new classifier from recovering it [10], and site can be decoded even by randomly initialized networks [11]. Similarity to the input, in turn, rewards doing nothing.

That retrained classifiers can still predict site after harmonization is known for ComBat [12], histogram matching [13] and HACA3 [14]. What has not been measured is how much probe protocols disagree on the same harmonized images. We audit this on ten outputs of seven methods, starting from a frozen probe: how much its verdict depends on training, whether low accuracy survives probes trained on the outputs, and what controls for head shape, intensity and target alignment add. We do not rank harmonizers; we derive a reporting checklist (Sec. 4).¹

2. DATA AND METHODS

Cohort. We use raw T1-weighted scans of 1,112 ABIDE I subjects from 17 sites [15], split by subject and stratified by site into 890 training, 113 validation and 109 test subjects. Each volume is clipped to its 1st–99th intensity percentiles and scaled to [0, 1]. We keep one axial slice per subject, the

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¹Code, protocol and per-subject results: <https://github.com/muqsitamir/HarmonIt>.

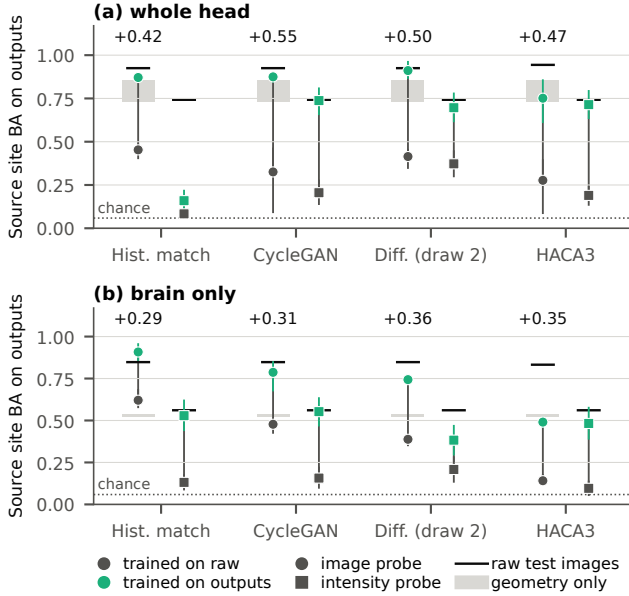


Fig. 1. Is site still decodable? Probes trained on raw slices (grey) or on a method’s outputs (aqua), all tested on that method’s outputs, for (a) whole heads and (b) brain-only slices. Circles: image probes (mean and range of three seeds); squares: intensity-histogram probes (95% interval). Numbers: gain of output-trained over raw-trained image probes. Black bars: raw-trained probe on raw slices; grey bands: probes that see only the head or brain outline.

one with the largest head area, cropped to the head and resized to 256×256 ; every method is evaluated on the same slices. NYU, the largest site, is the harmonization target, so we evaluate the 90 test subjects from the 16 other sites.

Harmonized outputs. We evaluate outputs of six methods implemented by us: NeuroCombat [1, 16], applied per pixel and fit on the test subjects themselves; histogram matching [17] to a reference pooled over all training sites; CycleGAN [18]; StarGAN [19] in an aggressive and a conservative setting; a DLEST-style model [6] without its energy-based sampler, after 1,000 and 1,500 training steps; and conditional diffusion translation [20, 21], sampled twice. HACA3 [22] is used with its authors’ pretrained weights and an NYU target image; its required bias correction and registration to MNI space are undone afterwards, so its outputs align with the original slices. Table 1 shows eight of the ten outputs. As a control, we also blur the raw test slices with a Gaussian filter, which changes the images without moving them toward NYU. Each analysis was fixed in a dated protocol before its results were seen.

Image change and target alignment. We compare each output with its input using PSNR and normalized cross-correlation (XCorr), and measure how much its intensity distribution changed with the Wasserstein-1 distance W . To

check whether outputs moved toward the target, we compare their intensity distribution with that of the 147 NYU training subjects (Wasserstein distance and KL divergence); ΔW_{NYU} is the output’s distance minus the input’s, so negative values mean closer to NYU.

Site probes. Image probes are ResNet-18 classifiers [23] that predict the site of a slice; we report balanced accuracy (BA) over the 16 source sites. We compare five families of probes:

Frozen: one probe trained once on raw images and applied unchanged to every output.

Same recipe: the frozen probe’s training recipe rerun with five new random seeds.

Converged: five probes trained four times longer, until validation accuracy is stable.

Shuffled labels: probes trained on permuted labels, giving the chance level actually reached (0.03–0.08; uniform guessing gives 0.06).

Slice probes: trained on one slice per training subject, either raw or a method’s outputs (three seeds). Control versions see only the head or brain outline, only intensity histograms, or only the brain (HD-BET masks [24]).

The first four are trained on raw images and tested on each method’s outputs (Table 1); slice probes compare raw-trained and output-trained probes on the same outputs (Fig. 1). Intervals are 95% bootstrap intervals, resampling test subjects within each site; they reflect these subjects and trained probes, not uncertainty across retraining or new cohorts.

3. RESULTS

Probes see more than scanner appearance. All probes recognized the site of raw test images well (BA 0.81–1.00). Head outlines alone already gave 0.73–0.86, whereas age, sex and diagnosis gave only 0.22, so site is visible in head geometry, not only in image contrast.

The verdict depends on how the probe is trained. Re-running the frozen probe’s recipe with new seeds gave very different accuracies on the same outputs (Table 1); for CycleGAN, anywhere between 0.13 and 0.59. Accuracy on raw images did not reveal this: one probe with raw BA 0.98 gave CycleGAN 0.20. Training longer made the probes agree, but they then found more site information than the frozen probe on every output except NeuroCombat and HACA3; for aggressive StarGAN, 0.65–0.81 instead of 0.21. The frozen probe’s verdict is thus one draw from an unstable process. Because the longer recipe changes several settings at once, it shows that the verdict is sensitive to training, not why.

A blur fools the frozen probe. The Gaussian blur kept PSNR higher than CycleGAN or diffusion (Fig. 2) and moved intensities away from NYU, yet it brought the frozen probe down to 0.07, the level of a probe trained on shuffled labels. The probe labelled almost every blurred image as one site,

Table 1. Results on the 90 source test subjects (means). Arrows give the direction usually read as better. Change: similarity to the input (PSNR, XCorr) and distance between input and output intensities (W). Target alignment: change in Wasserstein distance (ΔW_{NYU}) and KL divergence (KL_{NYU}) to NYU intensities. Site BA: balanced accuracy of probes trained on unharmonized images and tested on each row; frozen probe with 95% interval and share of outputs it labels NYU; same recipe and converged: mean (min–max) of five probes; blur: frozen probe only. [†]Fit on the test subjects. [‡]Reference pooled over all sites. [§]CSF rendered near zero.

Method	Change			Target alignment		Site BA: probes trained on unharmonized images, tested on each row			
	PSNR $\uparrow$	XCorr $\uparrow$	W	$\Delta W_{\text{NYU}}\downarrow$	$\text{KL}_{\text{NYU}}\downarrow$	Frozen probe $\downarrow$	$\rightarrow\text{NYU}\uparrow$	Same recipe $\downarrow$	Converged $\downarrow$
Raw input	–	–	0	0	0.30	0.95	0%	0.90 (0.81–0.98)	1.00 (1.00–1.00)
NeuroCombat [†]	20.0	0.940	0.055	+0.003	0.88	0.27 [0.21, 0.32]	2%	0.33 (0.26–0.37)	0.34 (0.28–0.42)
Histogram matching [‡]	23.0	0.982	0.054	–0.053	0.09	0.44 [0.37, 0.52]	10%	0.44 (0.27–0.59)	0.61 (0.55–0.68)
CycleGAN	21.3	0.977	0.048	–0.036	0.25	0.31 [0.25, 0.37]	52%	0.28 (0.13–0.59)	0.49 (0.44–0.54)
StarGAN (aggr.)	16.0	0.896	0.071	–0.019	0.32	0.21 [0.15, 0.27]	56%	0.35 (0.18–0.54)	0.70 (0.65–0.81)
DLEST-style 1000	28.4	0.993	0.017	–0.009	0.31	0.71 [0.65, 0.77]	6%	0.84 (0.76–0.89)	0.99 (0.96–1.00)
Diffusion, draw 1	21.7	0.948	0.008	–0.001	0.83	0.37 [0.32, 0.42]	36%	0.42 (0.24–0.63)	0.57 (0.52–0.59)
Diffusion, draw 2	21.7	0.948	0.008	–0.002	0.82	0.35 [0.29, 0.40]	38%	0.42 (0.30–0.62)	0.56 (0.50–0.59)
HACA3 (pretrained) [§]	18.9	0.954	0.062	–0.016	2.29	0.25 [0.18, 0.32]	40%	0.29 (0.18–0.37)	0.28 (0.23–0.33)
Blur control, $\sigma=2$	24.7	0.974	0.018	+0.016	0.57	0.07 [0.06, 0.10]	0%	–	–

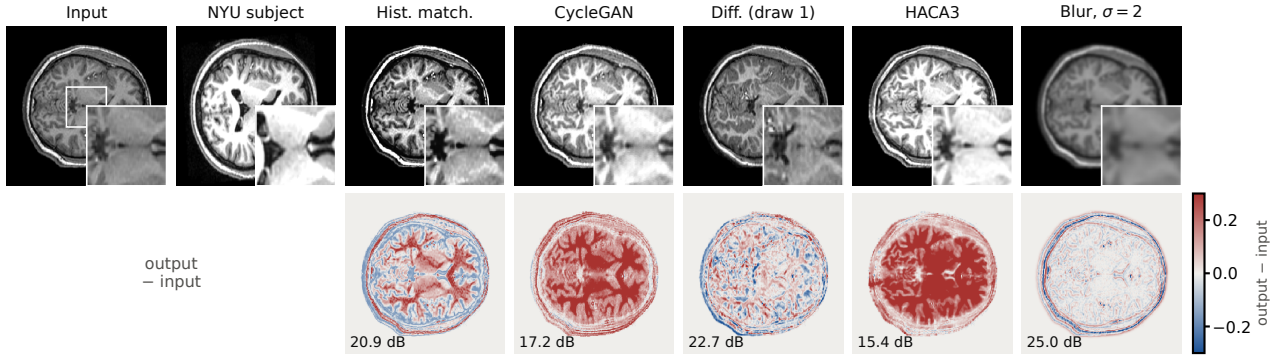


Fig. 2. One source subject (median PSNR) and an NYU subject with a similar slice, in the input’s intensity window, with 2 $\times$ zoomed insets. Bottom: output minus input, with PSNR. The blur has the highest PSNR of the outputs shown.

Yale, and none as NYU, whereas most of its errors on CycleGAN, aggressive StarGAN and diffusion outputs pointed to NYU. Checking where errors point, as PRISM does, therefore exposes the blur; as HACA3 shows below, it cannot reveal site information that remains in the outputs.

Site remains decodable after harmonization. Raw-trained slice probes dropped from 0.92 on raw slices to 0.33–0.45 on histogram-matched, CycleGAN and diffusion outputs, a drop that would usually be read as successful harmonization [9]. Probes trained on the outputs themselves, however, recovered the site at 0.87–0.91 (Fig. 1a). Because head shape alone identifies sites, we repeated this within the brain: output-trained probes still clearly outperformed raw-trained ones (Fig. 1b), and so did probes that see only intensity histograms. The only exception was histogram matching, which hid the site from whole-head histograms because it matches them by construction.

HACA3. The published model kept every probe trained on raw images low (frozen 0.25, retrained 0.28–0.29), and most of the frozen probe’s errors pointed to NYU. Yet probes

trained on its outputs recovered the site at 0.75, and its intensity histograms alone at 0.71. Within the brain, image probes did no better than brain shape alone, but intensity histograms still recovered the site (0.48 vs. 0.10 when trained on raw images). Part of HACA3’s effect came from its preprocessing: bias correction and registration alone lowered the frozen probe from 0.95 to 0.61, and HACA3 lowered it by a further 0.35. Five-fold cross-validation over all 928 source subjects (folds stratified by site, 10% of each training fold for model selection) gave the same picture (0.25 for raw-trained and 0.77 for HACA3-trained probes).

Small change is not target alignment. Diffusion changed intensities the least, yet closed only 1% of their distance to NYU and nearly tripled their KL divergence from it. Its second sample differed from the first more than from the input.

4. DISCUSSION

A single site-probe number is not a harmonization verdict: it changed by up to 0.46 between training seeds, was lowest for a blur, and missed the site information that HACA3's intensities kept. We therefore recommend reporting (1) several well-trained probes, with their recipe, their accuracy on raw images, a shuffled-label baseline and similarity to the input; (2) probes trained on the harmonized outputs, with controls for head shape and intensity, keeping in mind that high accuracy shows remaining site information, not by itself failed harmonization; (3) alignment with the target separately from the amount of change; and (4) uncertainty over subjects and over generator samples.

Limitations. We studied one cohort, one target site and one slice per subject, with 2–15 test subjects per source site. Without ground truth, PSNR and XCorr measure agreement with the input, not correctness. Site is entangled with demographics, anatomy and acquisition geometry; our controls bound these factors but cannot separate scanner effects from biological ones. Apart from HACA3, the methods are our own untuned implementations, so this is a focused counterexample rather than a survey.

5. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by ABIDE I [15], whose sites shared the fully anonymized data with local ethics approval or an explicit waiver. Ethical approval was not required as confirmed by the license attached with the open access data.

6. ACKNOWLEDGMENTS

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